

```

visited = []

start_state = [False, False, False, False]

goal_state = [True, True, True, True]

names = ['Man', 'Lion', 'Goat', 'Grass']

def validationFunction(current_state):

    if current_state[1] == current_state[2] and current_state[0] != current_state[1]:

        return False

    if current_state[2] == current_state[3] and current_state[0] != current_state[2]:

        return False

    else:

        return True

def stateGeneration(current_state):

    generated_states = []

    for position in range(0, len(current_state)):

        temp_state = current_state.copy()

        if position == 0:

            temp_state[0] = not temp_state[0]

            generated_states.append(temp_state)

        elif current_state[position] == current_state[0]:

            temp_state[0] = not temp_state[0]

            temp_state[position] = not temp_state[position]

            generated_states.append(temp_state)

    return generated_states

def dfsAlgo(current_state, visited):

    starting_side = []

    target_side = []

```

```
if current_state == goal_state:
    for index in range(0, len(current_state)):
        if current_state[index] == False:
            starting_side.append(names[index])
        if current_state[index] == True:
            target_side.append(names[index])
    print('starting side: ', starting_side)
    print('target side: ', target_side)
```

```
    return True
```

```
if not validationFunction(current_state):
    return False
```

```
if current_state in visited:
    return False
```

```
visited.append(current_state)
```

```
for index in range(0, len(current_state)):
    if current_state[index] == False:
        starting_side.append(names[index])
    if current_state[index] == True:
        target_side.append(names[index])
print('starting side: ', starting_side)
```

```
print('target side: ', target_side)
print("")
```

```
for child in stateGeneration(current_state):
```

```
    if dfsAlgo(child, visited):
```

```
        return 'Complete'
```

```
return False
```

```
dfsAlgo(start_state, visited)
```